

Analysis pipeline

8 × 10x filtered feature-barcode matrices · GEO GSE202212

ctrl 1–2, 3 dpMNU 1–2, 7 dpMNU 1–2, 10 dpMNU 1–2 · GRCz11 + EGFP

Load and annotate metadata

step 01 · 20,097 cells, 25,433 features

Quality control

step 02 · ≥ 200 genes, $< 15\%$ mito $\rightarrow$ 15,893 cells (79.1%)

Normalise, HVG, scale

step 03 · $1e4$ target, 2,000 HVGs, 2 covariates regressed

PCA, neighbours, UMAP, Leiden

step 04 · 20 PCs, $k=15$, resolution 0.6 $\rightarrow$ 20 clusters

Marker discovery and annotation

step 05 · Wilcoxon on all 22,813 genes $\rightarrow$ 13 cell types

Reporter dynamics

step 06 · §3.5–3.7
EGFP within 0.2 pp
of published

Müller glia states

step 06 · §3.8–3.9
resting vs activated
23/23 signature genes

Photoreceptors

step 06 · §3.10–3.11
rods not reproduced
cones reproduced

Published panels

step 07 · §4.1
14 of 18 reproduced

Sensitivity analyses · steps 08 and 09 · §4.7 and §4.8

Harmony correction and clustering at the published resolution: neither recovers the rod split; the reporter time course is unchanged to two decimal places